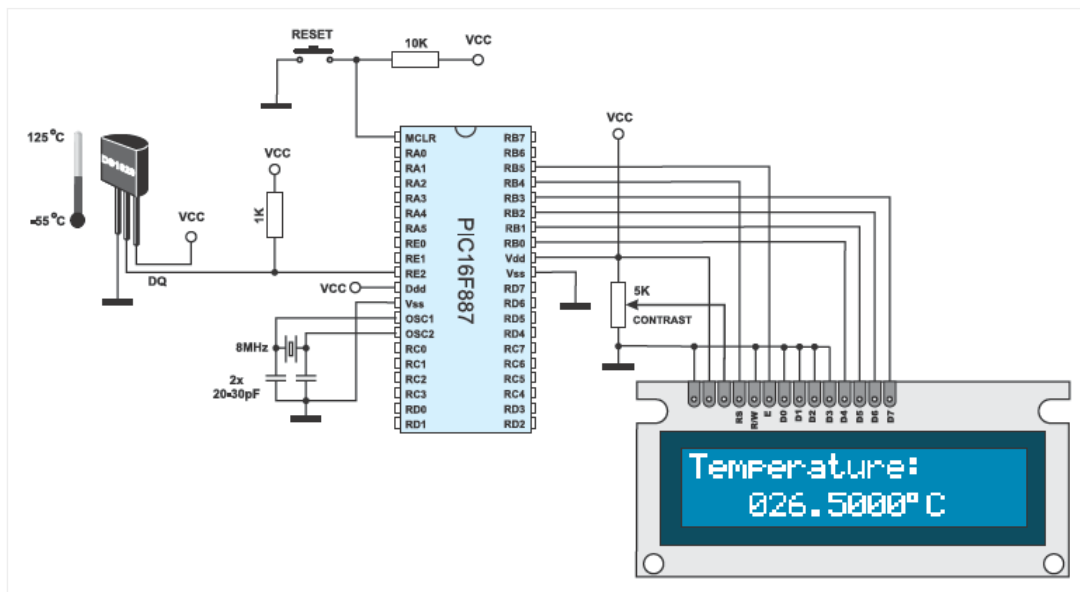


4.14 EXAMPLE 12

MIKROELEKTRONIKA

Temperature measurement using DS1820 sensor. Use of '1-wire' protocol...

Temperature measurement is one of the most common tasks performed by the microcontroller. A DS1820 sensor is used for measurement here. It is capable of measuring temperature in the range of -55 °C to 125 °C with 0.5 °C accuracy. For the purpose of transferring data to the microcontroller, a special type of serial communication called *1-wire* is used.



Due to a simple and wide use of these sensors, commands used to run and control them are in the form of functions stored in the One_Wire library. There are three functions in total:

- **Ow_Reset** is used for resetting sensor;
- **Ow_Read** is used for receiving data from sensor; and
- **Ow_Write** is used for sending commands to sensor.

This example implies the advantage in using libraries with ready-to-use functions. Concretely, you don't have to study documentation provided by the manufacturer in order to use this sensor. It is sufficient to copy some of these functions in the program. If you want to know how any of them is declared, just right click on it and select the *Help* option.

```
1  /*Header*****//
2
3  // LCD module connections
4  sbit LCD_RS at RB4_bit;
5  sbit LCD_EN at RB5_bit;
6  sbit LCD_D4 at RB0_bit;
7  sbit LCD_D5 at RB1_bit;
8  sbit LCD_D6 at RB2_bit;
9  sbit LCD_D7 at RB3_bit;
10 sbit LCD_RS_Direction at TRISB4_bit;
```

```

11 sbit LCD_EN_Direction at TRISB5_bit;
12 sbit LCD_D4_Direction at TRISB0_bit;
13 sbit LCD_D5_Direction at TRISB1_bit;
14 sbit LCD_D6_Direction at TRISB2_bit;
15 sbit LCD_D7_Direction at TRISB3_bit;
16 // End LCD module connections
17
18 const unsigned short TEMP_RESOLUTION = 9;
19 char *text = "000.0000";
20 unsigned temp;
21
22 void Display_Temperature(unsigned int temp2write) {
23     const unsigned short RES_SHIFT = TEMP_RESOLUTION - 8;
24     char temp_whole;
25     unsigned int temp_fraction;
26
27     // check if temperature is negative
28     if (temp2write & 0x8000) {
29         text[0] = '-';
30         temp2write = ~temp2write + 1;
31     }
32     // extract temp_whole
33     temp_whole = temp2write >> RES_SHIFT ;
34
35     // convert temp_whole to characters
36     if (temp_whole/100)
37         text[0] = temp_whole/100 + 48;
38     else
39         text[0] = '0';
40
41     text[1] = (temp_whole/10)%10 + 48; // Extract tens digit
42     text[2] = temp_whole%10 + 48; // Extract ones digit
43
44     // extract temp_fraction and convert it to unsigned int
45     temp_fraction = temp2write << (4-RES_SHIFT);
46     temp_fraction &= 0x000F;
47     temp_fraction *= 625;
48
49     // convert temp_fraction to characters
50     text[4] = temp_fraction/1000 + 48; // Extract thousands digit
51     text[5] = (temp_fraction/100)%10 + 48; // Extract hundreds digit
52     text[6] = (temp_fraction/10)%10 + 48; // Extract tens digit
53     text[7] = temp_fraction%10 + 48; // Extract ones digit
54
55     // Display temperature on LCD
56     Lcd_Out(2, 5, text);
57 }
58
59 void main() {
60     ANSEL = 0; // Configure AN pins as digital I/O
61     ANSELH = 0;
62     C1ON_bit = 0; // Disable comparators
63     C2ON_bit = 0;
64
65     Lcd_Init(); // Initialize LCD
66     Lcd_Cmd(_LCD_CLEAR); // Clear LCD
67     Lcd_Cmd(_LCD_CURSOR_OFF); // Turn the cursor off
68     Lcd_Out(1, 1, " Temperature: ");
69
70     // Print degree character, 'C' for Centigrades
71     Lcd_Chr(2,13,223); // different LCD displays have different char code for degree
72     // if you see greek alpha letter try typing 178 instead of 223
73
74     Lcd_Chr(2,14,'C');
75
76     //--- main loop
77     do {
78         //--- perform temperature reading
79         Ow_Reset(&PORTE, 2); // Onewire reset signal
80         Ow_Write(&PORTE, 2, 0xCC); // Issue command SKIP_ROM
81         Ow_Write(&PORTE, 2, 0x44); // Issue command CONVERT_T
82         Delay_us(120);
83         Ow_Reset(&PORTE, 2);
84         Ow_Write(&PORTE, 2, 0xCC); // Issue command SKIP_ROM
85         Ow_Write(&PORTE, 2, 0xBE); // Issue command READ_SCRATCHPAD
86         temp = Ow_Read(&PORTE, 2);
87         temp = (Ow_Read(&PORTE, 2) << 8) + temp;
88
89         //--- Format and display result on Lcd
90         Display_Temperature(temp);
91         Delay_ms(500);
92     } while (1);
93 }

```

In order to make this example work properly, it is necessary to tick off the following libraries in the Library Manager prior to compiling:

- One_Wire
- LCD



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